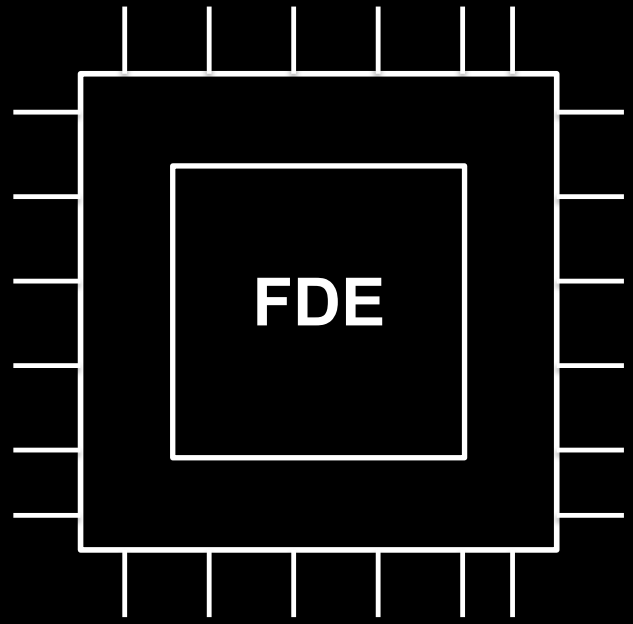
COMPUTER HARDWARE IN ACTION

buses • cloud & NAS • embedded systems

Pearson Edexcel International GCSE Computer Science
Unit 4: Computers — Hardware | Student Book (2020)



FETCH / DECODE / EXECUTE

Buses: pathways for communication

Each bus carries a different kind of information between the CPU, memory and devices.

ADDRESS BUS

Carries the address of the memory location or device being accessed.

Usually one-way: CPU to memory / I/O.

It identifies where; it does not carry the data itself.

DATA BUS

Carries instructions and data between components.

Usually two-way: CPU to and from memory / I/O.

The bus carries the actual value or instruction.

CONTROL BUS

Carries command and status signals, such as READ or WRITE.

Signals can travel in different directions, depending on their purpose.

DURING A FETCH **Address bus: address • Control bus: READ signal • Data bus: instruction returns to the CPU**

Contemporary storage: cloud and NAS

The physical data still resides on storage devices; the access model is different.

CLOUD STORAGE

WHAT IT IS
Files are held on a provider's remote servers and accessed over the internet.

WHY USE IT
Access from different places/devices; easy sharing and capacity changes.

TRADE-OFFS
Internet or service outages can block access; consider privacy, security and ongoing costs.

NETWORK-ATTACHED STORAGE (NAS)

WHAT IT IS
A dedicated storage device connected to a local network and shared by users/devices.

WHY USE IT
Convenient local file sharing, with more control over where files are kept.

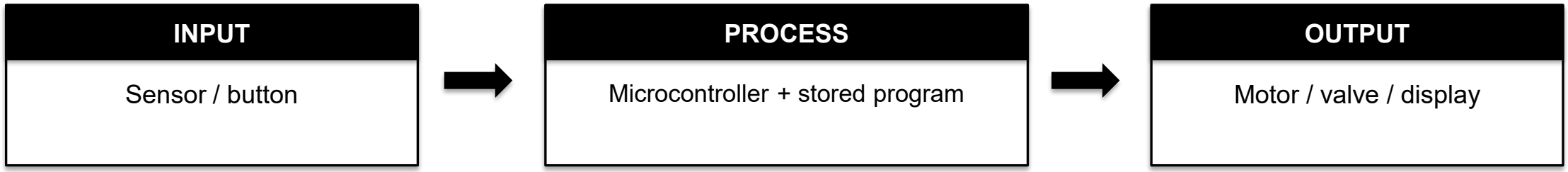
TRADE-OFFS
The owner maintains the hardware, access controls and backups; remote access needs extra setup.

Both are secondary storage. Choose based on access, capacity, reliability, privacy/security and cost.

Embedded systems: computers inside products

A dedicated computer controls one product or a narrow set of functions.

Used when a product needs compact, automatic control of a specific task — rather than a general-purpose computer.



WASHING MACHINE

Sensors report conditions; the controller operates the motor, pump and heater through the cycle.

CAR ABS

Wheel-speed sensors feed a controller that adjusts braking when a wheel is about to lock.

THERMOSTAT

A temperature sensor feeds a controller that switches heating on or off to meet a target.

Same pattern: input is sensed → a stored program decides → an output is controlled.